

Roberto Alarcon

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Professional Summary

Performance-obsessed Senior Software Engineer with 17+ years in C#/.NET, distributed systems, and event-driven architecture. Proven track record resolving critical production incidents, mitigating database storms, and optimizing high-throughput Kafka and PostgreSQL pipelines. Full-stack across .NET Core, Angular, React, and Azure – now integrating AI capabilities via OpenAI SDK and Claude Code.

Technical Skills

- Backend: .NET Core, C#, PostgreSQL, NATS, Kafka, Go, Redis, Python, SQL Server, Nodejs
- Frontend: Angular, React, TypeScript, JavaScript, Tailwind
- AI/ML: OpenAI SDK, Claude Code, PyTorch, scikit-learn
- Infrastructure: Kubernetes, Docker, Azure, Linux, Functions, windows server

Experience

Senior Software Engineer, Calsense, San Diego, CA (2023 - Present)

- System Architecture & Performance: Acted as sole engineer managing an IoT backend supporting hundreds of thousands of concurrent devices; scaled system uptime from 80% to 99%.
- Database Storm Mitigation: Resolved critical database storms and bottlenecked relational storage by introducing NATS to queue and throttle heavy incoming request volumes.
- Concurrency & Reliability: Identified and resolved thread starvation bugs by implementing native .NET synchronization primitives (SemaphoreSlim) to protect shared database connection pools.
- AI Feature Innovation: Integrated the OpenAI SDK to implement automated intelligent reporting features, translating complex system analytics into natural language insights for users.

- Database Migration: Spearheaded a 9-week zero-downtime migration of a legacy on-premise database to Azure PostgreSQL, strictly maintaining interface contracts for legacy client applications.
- Data Access Optimization: Cut average UI load times by 50% for high-volume enterprise clients by redesigning data structures, batching queries, rewriting raw SQL, and tuning indexes.
- Event-Driven Messaging: Engineered an internal pub/sub messaging architecture utilizing a partitioned Kafka cluster to balance message throughput and increase processing efficiency.
- Full-Stack Scope: Deliver software across the entire stack, utilizing C for embedded IoT logic, .NET Core/Python for backend services, and Angular/React for front-end portals.

Skills: C#, Python, JavaScript, Postgresql, Kafka, Pandas, PyTorch, OpenAIsDK

Software Engineer III, Lytx INC, San Diego, CA (2023)

- I lead the design of a new version of the public API for customers to access their data.
- Added features to existing portals for customers.
- Fixed performance bugs in an existing application.

Skills: C#, Kafka, Redis, SQL Server, JavaScript, MongoDB

Senior Software Engineer, Becton Dickinson, San Diego, CA (2015 - 2022)

- Led a full-scale architectural refactor of a legacy Windows application into a native Android application for specialized medical scanning hardware.
- Built end-to-end REST APIs and microservices infrastructure utilizing Docker, Kubernetes, and Kafka pipelines to support cross-functional data consumption.
- Enhanced various web applications using React, Vue, and legacy JavaScript frameworks (Knockout.js, Backbone.js).

Skills: Java, Python, C#, .NET Core, Flask, Postgresql, RabbitMQ, Pytorch

Software Engineer II, Innovo Commerce, Irvine, CA (2013 - 2015)

- Led the front-end team to design, build, and deploy enterprise clinical trial applications for major pharmaceutical clients.

Skills: Knockout.js, JavaScript, C#, Neo4J, Apache Cordova, SQL Server, CSS

Lead Front-End (co-founder), Clustter Technologies, Remote (2010 - 2015)

- Co-founded a real-time web startup handling team communication, messaging, and task management for 50+ enterprise customers.

Skills: Python, JavaScript, BackboneJS, Pylons, Postgresql, SQLAlchemy.

Education Bachelor of Science in Computer Science (equivalent), CETYS University, Tijuana, Mexico 2005 - 2011